

(RATIO & PROPORTION)

1. The sum of three numbers is 98. If the ratio of the first to the second is 2:3 and that of second to the third is 5:8 then the second number is?
 - a. 20
 - b. 30
 - c. 38
 - d. 48
 - e. 52

2. Rs. 73689 are divided between A and B in the ratio 4:7. What is the difference between thrice the share of A and twice the share of B?
 - a. Rs. 36699
 - b. Rs. 46893
 - c. Rs. 20097
 - d. Rs. 26796
 - e. Rs. 13398

3. The total number of students in a school is 31700. If the ratio of boys to the girls in the school is 743:842 respectively, what is the total number of girls in the school?
 - a. 14860
 - b. 16480
 - c. 15340
 - d. Cannot be determined
 - e. None of these

4. How much is C's share if Rs. 10,980 is to be divided amongst A, B, and C in the ratio 7:3:5 respectively?
 - a. Rs. 3,600
 - b. Rs. 3,006
 - c. Rs. 3,650
 - d. Rs. 3,660
 - e. Rs. 3,124

5. What is the sum of B's and C's share?
 - a. Rs. 5,685

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- b.** Rs. 5,865
- c.** Rs. 5,897
- d.** Rs. 5,873
- e.** Rs. 5,856

6. What is the difference between A's and B's shares?

- a.** Rs. 2,196
- b.** Rs. 2,928
- c.** Rs. 2,961
- d.** Rs. 2,289
- e.** Rs. 2,982

7. A sum of Rs. 221 is divided among X, Y, and Z such that X gets Rs. 52 more than Y. Y gets Rs. 26 more than Z. The ratio of the shares of X, Y, and Z respectively is:

- a.** 9:5:3
- b.** 9:3:5
- c.** 5:9:3
- d.** 10:6:5
- e.** None of these

8. If 50% of a certain number is equal to $\frac{3}{4}$ th of another number, what is the ratio between the numbers?

- a.** 3:2
- b.** 2:5
- c.** 5:2
- d.** 3:4
- e.** None of these

9. The ratio of the earnings of A and B is 4:7 respectively. If the earnings of A increase by 50% and the earnings of B decrease by 25%, the new ratio of their earnings becomes 8:7 respectively. What are A's earnings?

- a.** 26000
- b.** 28000
- c.** 21000

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- d. Data inadequate
- e. None of these

10. The cost of making an article is divided between materials, labour, and overheads in the ratio of 3:4:1. If the material costs Rs. 234, then the labour cost is:

- a. Rs. 176
- b. Rs. 312
- c. Rs. 78
- d. Rs. 390
- e. None of these

11. The ages of Mira, Tina, and Sania are in the ratio of 6:4:7 respectively. If the sum of their ages is 34 years, what is Sania's age?

- a. 12 yr.
- b. 10 yr.
- c. 18 yr.
- d. 8 yr.
- e. None of these

12. In a school, the number of boys and girls are in the respective ratio of 2:3. If the number of boys is increased by 20% and that of girls is increased by 10%, what will be the new ratio of the number of boys to girls?

- a. 14:5
- b. 5:8
- c. 13:4
- d. Data inadequate
- e. None of these

13. When x is subtracted from the numbers 9, 15, and 27, the remainders are in continued proportion. What is the value of x ?

- a. 8
- b. 6
- c. 4

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- d. 5
- e. None of these

14. The price of sugar is increased by 20%. If the expenditure is not allowed to increase, the ratio between the reduction in consumption and the original consumption is:

- a. 1:3
- b. 1:4
- c. 1:6
- d. 1:5
- e. None of these

15. The ratio between two numbers is 2:3. If each number is increased by 4, the ratio between them becomes 5:7. What is the difference between the numbers?

- a. 8
- b. 6
- c. 4
- d. 2
- e. None of these

16. Seats for Mathematics, Physics, and Biology in a school are in the ratio 5:7:8. There is a proposal to increase these seats by 40%, 50%, and 75% respectively. What will be the ratio of the increased seats?

- a. 2:3:4
- b. 6:7:8
- c. 6:8:9
- d. Cannot be determined
- e. None of these

17. The salaries A, B, and C are in the ratio 2:3:5. If their salaries were increased by 15%, 10%, and 20% respectively, what will be the new respective ratio of their salaries?

- a. 3:3:10
- b. 23:33:60
- c. 10:11:20

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- d. Cannot be determined
- e. None of these

18. Production of company A is 120% of the production of company B and 80% of the production of company C. What is the ratio between the productions of companies A, B, and C respectively?

- a. 6:5:9
- b. 6:5:4
- c. 12:10:15
- d. 10:12:15
- e. None of these

19. The ratio between two numbers is 4:3, and their L.C.M. is 264. The second number is:

- a. 66
- b. 44
- c. 55
- d. 88
- e. None of these

20. Two numbers are respectively 30% and 20% more than a third number. Find the ratio of the two numbers.

- a. 12:13
- b. 13:12
- c. 3:2
- d. 2:3
- e. None of the above

21. What number has to be added to each term of 4:7 to make the ratio 5:6?

- a. 13
- b. 12
- c. 10
- d. 11
- e. None of these

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22. If $a:b = 9:5$ and $b:c = 7:4$, then $a:b:c = ?$

- a. 14:10:17
- b. 35:63:20
- c. 63:35:20
- d. 20:36:63
- e. None of the above

23. In the 45 litres mixture of milk and water, the ratio of milk and water is 5:4. Find the quantity of water required to be added so that the resultant mixture will be in the ratio 4:5.

- a. 7.75 litres
- b. 11.25 litres
- c. 9.25 litres
- d. 12.50 litres
- e. None of these

24. Two natural numbers are in the ratio of 4:7 and their product is 112. Find both the numbers.

- a. 4 and 7
- b. 8 and 14
- c. 12 and 21
- d. 16 and 28
- e. None of these

25. A starts a business with Rs.3500. After 5 months, B joins with A as his partner. After one year the profit is divided in the ratio 2:3. What is B's contribution in capital?

- a. Rs.8000
- b. Rs.8500
- c. Rs.9000
- d. Rs.7500
- e. None of these

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26. The monthly income of A and B is in the ratio of 4:3 and their monthly expenditure is in the ratio of 3:2. If each of them saves Rs.6000 per month, the income of B is:

- a.** 12000
- b.** 24000
- c.** 18000
- d.** 36000
- e.** None of these

27. If you are asked to divide Rs. 13950 among three of your friends, such that the 2nd friend should get double of the 1st friend and the 3rd friend should get Rs. 50 less than the double of the 2nd friend's share, how much will you have to give to your 1st friend?

- a.** Rs.2010
- b.** Rs.2050
- c.** Rs.1950
- d.** Rs.2000
- e.** None of these

28. The fourth proportional to 8, 12, and 16 is:

- a.** 24
- b.** 32
- c.** 20
- d.** None of these

29. Incomes of two companies A and B are in the ratio of 5:8. Had the income of company A been more by Rs.25 lakh, the ratio of their incomes would have been 5:4. What is the income of company B?

- a.** Rs.80 lakh
- b.** Rs.50 lakh
- c.** Rs.40 lakh
- d.** Rs.60 lakh
- e.** None of these

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30. Salaries of A, B, and C were in the ratio 3:5:7 respectively. If their salaries were increased by 50%, 60%, and 50% respectively, what will be the new ratio of their respective salaries?

- a. 3:6:7
- b. 4:5:7
- c. 4:5:8
- d. Data inadequate
- e. None of these

31. Rs. 558 is divided among three brothers Sanju, Birju, and Makhon such that 12 times Sanju's share is equal to 6 times Birju's share which is equal to 8 times Makhon's share. How much Sanju got?

- a. Rs.124
- b. Rs.248
- c. Rs.224
- d. Rs.186
- e. None of these

32. Salaries of Rajesh and Sunil are in the ratio of 2:3. If the salary of each one is increased by Rs.4000, the new ratio becomes 40:57. What is Sunil's present salary?

- a. Rs.17000
- b. Rs.20000
- c. Rs.25500
- d. Cannot be determined
- e. None of these

33. In the squadron of Indian Air Force the ratio of Sukhoi to Mig and Jaguar together is 5:7 and the ratio of Jaguar to Sukhoi and Mig together is 1:2. Find the ratio of Sukhoi to Mig?

- a. 2:7
- b. 3:5
- c. 3:1

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- d. 5:3
- e. None of these

34. The number of pens in three different pencil boxes is in the ratio of 1:2:3. Find the ratio in which the number of pens in the first and the second boxes must be increased so that the new ratio becomes 3:2:1.

- a. 1:3
- b. 2:1
- c. 2:3
- d. 3:4
- e. None of these

35. The sum of three numbers is 98. If the ratio between the first and second be 2:3 and that between the second and the third be 5:8, then find the second number.

- a. 42
- b. 50
- c. 35
- d. 30
- e. None of these

36. A man spends his two months' income in three months' time. If his monthly income is Rs. 6000, then his annual saving is:

- a. Rs.18000
- b. Rs.24000
- c. Rs.12000
- d. Rs.36000
- e. None of these

37. The sum of $\frac{1}{5}$ th of the number and 25% of another number is equal to 40% of the first number. What is the ratio of the first number and the second number?

- a. 4:3
- b. 5:2

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- c. 5:4
- d. 6:5

38. A bucket contains a mixture of two liquids A & B in the proportion 5:3. If 16 litres of the mixture is replaced by 16 litres of liquid B, then the ratio of the two liquids becomes 3:5. How much of the liquid B was there in the bucket?

- a. 16.5 l
- b. 18 l
- c. 14.5 l
- d. 15 l
- e. None of these

39. In a college, the number of students studying Arts, Commerce, and Science are in the ratio of 3:5:8 respectively. If the number of students studying Arts, Commerce, and Science is increased by 20%, 40%, and 25% respectively, what will be the new ratio of students in Arts, Commerce, and Science respectively?

- a. 4:8:5
- b. 3:10:10
- c. 4:18:5
- d. 32:35:25
- e. None of these

40. The students in three classes are in the ratio 4:6:9. If 12 students are increased in each class, the ratio changes to 7:9:12. Then the total number of students in the three classes before the increase is?

- a. 95
- b. 76
- c. 100
- d. 114
- e. None of these

41. Some apples are to be distributed among some boys. If two more

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apples are available, each boy gets six apples. If five less apples are available, each boy gets five apples. How many apples are available?

- a. 30
- b. 40
- c. 45
- d. 50
- e. None of these

42. Reena and Shaloo are partners in a business. Reena invests Rs. 35,000 for 8 months and Shaloo invests Rs.42,000 for 10 months. Out of a profit of Rs.31,570, Reena's share is:

- a. Rs.9471
- b. Rs.12,628
- c. Rs.18,040
- d. Rs.18,942
- e. None of these

43. A and B enter into a partnership and invested Rs. 1400 and Rs. 1800 respectively. If they split half of the profit equally for their efforts and the remaining balance in the ratio of their investments and B got Rs. 47 more than A, what was the profit made by the company?

- a. Rs.376
- b. Rs.652
- c. Rs.752
- d. Rs.954
- e. Rs.854

44. A starts a business with Rs.3500 and after 5 months, B joins with A as his partner. After a year, the profit is divided in the ratio 2:3. What is B's contribution in the capital?

- a. Rs.7500
- b. Rs.8000
- c. Rs.8500
- d. Rs.9000
- e. None of these

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45. A started a business with capital of Rs. 20000. Four months later, B joined as a partner with a capital of Rs. 10000. What is the share of A out of the total profit of Rs. 4000 at the end of the year?

- a. Rs.5000
- b. Rs.1200
- c. Rs.3000
- d. Rs.1800
- e. Rs.2850

46. Three partners started a business with Rs. 80000. At the end of the year, they receive Rs. 1800, Rs.3000, and Rs.4800 as profit. Find the investment of the second person.

- a. Rs.27000
- b. Rs.40000
- c. Rs.15000
- d. Rs.32000
- e. None of these

47. M and N start a business with Rs. 5000 and Rs. 7000 respectively. After 4 months, O joins the business with Rs. 9000. At the end of the year, O gets Rs. 1800 as his share of profit. Then find the difference between profit got by N and M?

- a. Rs.600
- b. Rs.300
- c. Rs.1200
- d. Rs.1500
- e. Rs.1250

48. A and B started a joint business. Investment of A was six times that of B and tenure of investment by A was also four times that of B. If B got a profit of Rs. 8000, then the total profit was?

- a. Rs.240000
- b. Rs.160000
- c. Rs.200000

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- d.** Rs.280000
- e.** Rs.245000

49. A and B established a firm together. A's investment was thrice that of B's. A also kept the investment for twice as much time as B. If B got a profit of Rs.4000, what was the total profit?

- a.** Rs.30,000
- b.** Rs.28,000
- c.** Rs.40,000
- d.** Rs.45,000
- e.** None of these

50. Riya and Sima invested in a partnership business. Riya invests Rs. 70,000 for 8 months and Sima invests Rs. 84,000 for 10 months. Out of a profit of Rs. 63140, Riya's share is:

- a.** Rs.25000
- b.** Rs.25256
- c.** Rs.24500
- d.** Rs.25270
- e.** None of these

OFFICIAL ANSWER KEY

Subject: Ratio & Proportion

Q.No	Ans	Q.No	Ans
1	(c)	26	(c)
2	(e)	27	(d)
3	(b)	28	(a)
4	(d)	29	(a)
5	(b)	30	(c)
6	(b)	31	(a)
7	(a)	32	(b)
8	(d)	33	(b)
9	(d)	34	(b)
10	(b)	35	(d)
11	(a)	36	(a)
12	(b)	37	(b)
13	(c)	38	(b)
14	(b)	39	(d)
15	(a)	40	(c)
16	(c)	41	(b)
17	(c)	42	(b)
18	(a)	43	(c)
19	(a)	44	(b)
20	(a)	45	(c)
21	(c)	46	(a)
22	(c)	47	(a)
23	(b)	48	(c)
24	(b)	49	(b)
25	(c)	50	(b)